Russell W. Mercer

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EDUCATION

Brigham Young University, Provo, UT January 2024 – April 2028 (Expected)

B.S. in Biochemistry, Minor in Computer Science, with machine learning courses

- Cumulative GPA 3.99; Science GPA 3.99
- Advanced biochemistry, pre-medical, and machine learning coursework

Medical College Admission Test (MCAT): 519, 96th percentile August 2026

RESEARCH EXPERIENCE

Biochemistry Research Assistant, Jensen Lab, BYU August 2026 – Present

Department of Chemistry and Biochemistry. Mentor: Grant J. Jensen, PhD, Dean, College of Physical and Mathematical Sciences; with Chitral Chatterjee

- Scoped and proposed two cryo-electron tomography projects, approved by the PI and awaiting institutional biosafety review: the surface architecture of *Bartonella henselae*, and the apical complex of *Babesia duncani*
- Testing whether a general-purpose AI coding agent can shorten the cryo-electron tomography learning curve for a new researcher without replacing the understanding the work normally builds
- Processing 39 tilt series of cryo-FIB-milled, exponentially growing *E. coli* through motion correction, alignment, reconstruction, missing-wedge correction, template matching, classification, and subtomogram averaging, ending in a ribosome neighbor analysis
- Recording which agent recommendations are accepted, overridden, or caught as errors, and packaging the workflow as a reusable agent skill

Clinical Research Assistant, UT Southwestern Medical Center May 2026 – Present

Radiation Oncology, Kenneth Westover, MD, PhD (May to August 2026); Medical Physics, Jie Deng, PhD (August 2026 to present)

- Started through the BYU Summer Premedical Research Internship, funded by the BYU Simmons Center for Cancer Research; on site in Dallas, TX through August 2026 and remote since
- Four studies: lung and pancreas diffusion-weighted MRI on the Elekta Unity MR-Linac, biology-guided radiotherapy on RefleXion, and radiolucent patient immobilization. First author on three manuscripts, one submitted and two in final co-author circulation, with more in preparation
- Rebuilt the department's diffusion MRI processing pipeline into a web application with a bulk runner, viewer, threshold tool, and fit checker; ran 76 patients in about 2.5 days of server time, and hospital IT moved it to the Radiation Oncology Application Team for clinical use
- Wrote the C#, Python, and R extraction and figure pipelines behind the studies; abstracted clinical data in Epic; wrote the hand-off document and trained the postdoc who inherited the pipeline
- Offered a continuing position in Medical Physics after the summer internship

Chemistry Research Assistant, Stowers Lab, BYU April 2024 – May 2026

Department of Chemistry and Biochemistry. Mentor: Kara J. Stowers, PhD

- Synthesized surface-bound metal-organic framework thin films (CuBDC/DABCO, HKUST-1) by spin coating and studied their structure and decomposition
- Helped build, repair, and maintain a flow reactor and an ultra-high-vacuum chamber reaching about 10^{-10} Torr, with X-ray photoelectron spectroscopy and mass spectrometry
- Trained newer lab members in spin coating and SEM imaging
- Supported by a BYU Undergraduate Research Award; still contributing to a manuscript in preparation

PUBLICATIONS

Peer-Reviewed

Ralph AD, **Mercer RW**, Hall SB, Butikofer BC, Collins CS, Sturgill CJ, Minson PS, Stefik M, Stowers KJ. Ambient pressure fabrication of $Cu_xZn_{1-x}(BDC)_2DABCO$ thin films through spin-coating and their temperature-dependent structural development. *Langmuir*. 2026;42(6):4475–4488. doi:10.1021/acs.langmuir.5c05182

Submitted

Mercer RW, et al. Onboard intravoxel incoherent motion diffusion-weighted MRI tracks early tumor-volume response in ultra-central lung tumors treated with PULSAR. UT Southwestern. Submitted to the *International Journal of Radiation Oncology, Biology, Physics*, October 2026.

In Co-Author Circulation

Mercer RW, et al. Dosimetric cost of a smaller target when using biology-guided radiation therapy for lung cancer. UT Southwestern.

Mercer RW, et al. Radiolucent stereotactic body frame for online adaptive radiation therapy. UT Southwestern.

Benefits of weekly on-demand adaptive radiation therapy for locally advanced lung cancer in responders. Co-author. UT Southwestern.

Online adaptive radiation therapy for early-stage lung SABR. Co-author. UT Southwestern.

In Preparation

Mercer RW, et al. Pancreas diffusion-weighted MRI on the Elekta Unity MR-Linac. UT Southwestern.

Mercer RW, Chatterjee C, Jensen GJ. Using an AI coding agent to learn cryo-electron tomography processing. Jensen Lab, BYU.

Co-author, surface chemistry in the Stowers Lab ultra-high-vacuum chamber. BYU.

Author order on manuscripts still in review is not final.

PRESENTATIONS AND ABSTRACTS

Mercer RW, Chatterjee C, Jensen GJ. Can an AI agent collapse the cryo-ET learning curve? Poster, Utah CryoEM Symposium, Brigham Young University, Provo, UT. — October 2026

Invited guest talk on UT Southwestern research and AI in research. HL 385R, Healthcare Research Hub, BYU Healthcare Leadership, Marriott School of Business, Provo, UT. — October 2026

UT Southwestern radiation oncology research. Poster, BYU Medical Society, Provo, UT. — August 2026

UT Southwestern radiation oncology research. Oral presentation, BYU Simmons Center for Cancer Research, Provo, UT. — August 2026

Metal-organic framework thin films. American Chemical Society Spring Meeting, Atlanta, GA. — March 2026

Metal-organic framework thin films. BYU Student Research Conference, Provo, UT. — February 2026

Fast room temperature synthesis and characterization of a surface bound metal organic framework without the use of a self-assembling monolayer. BYU Student Research Conference, Provo, UT. — March 2025

HONORS, AWARDS, AND FUNDING

SCCR Fellow, BYU Simmons Center for Cancer Research (funded summer research at UT Southwestern) — 2026

Outstanding Analytical Chemist Award, BYU — 2025

Dean's List, BYU (Winter, Spring, and Fall 2024; Fall 2025) — 2024 – 2025

Undergraduate Research Award, BYU (awarded each semester) — 2024 – 2026

Stoney Black Flight of the Phoenix Award, Air Academy High School Marching Band — 2021

Eagle Scout, Boy Scouts of America — 2020

CLINICAL EXPERIENCE

Medical Assistant, Health Plus Clinic, Springville, UT July 2025 – April 2026

- Assisted a D.O. and a P.A. with patient care and clinical procedures
- Kept medical records accurate and compliant; completed HIPAA and medical assistant training

Volunteer, Utah State Hospital, Provo, UT May 2025 – Present

- Physical Therapy (May 2025 to present): guide 10+ mental health patients through therapeutic exercise and help the physical therapist keep sessions safe and encouraging
- Pediatric Center (September 2025 to January 2026): led group activities for pediatric mental health patients
- Pediatric Mentor (December 2025 to May 2026): met one-on-one with pediatric mental health patients

PHYSICIAN SHADOWING

Steven M. Grover, Robotic Gynecologic Surgery, Utah Valley Hospital, 3.5 hours — October 2026

Thomas Fowler, Allergy, 4 hours — June 2026

Stanford C. Taylor, Retina Surgery, Pueblo, CO, 6 hours — April 2026

Matheson Harris, Plastic Surgery (Facial Reconstruction), Utah Surgical, 7.5 hours — November 2025

Michael Benson, Psychiatry, Utah State Hospital, 8 hours — August 2025

Dan Baer, Pain Management (PM&R), 3.5 hours — July 2025

James Barrott, General Surgery; Joshua Jaramillo, Trauma Surgery and ICU; Eric Cannon, Anesthesia. Scheduled. — November – December 2026

LEADERSHIP AND TEACHING

Officer, BYU AI Association September 2026 – Present

- Lead the weekly AI in Research meeting
- Speak about AI in research to other colleges across the university, and meet with professors and labs about using it in their own work
- Represented the association at BYU's AI for Social Innovation Conference, on the Data Sovereignty and Inclusion working table

Pre-Med Peer Mentor, BYU Pre-Med Association October 2026 – Present

- Mentor pre-med students on coursework, research, and clinical experience

Lead Program Director, BYU Y-Serve, Early Learning Essentials August 2024 – Present

- Program Director from August 2024; Lead Program Director since March 2025
- Recruit volunteers, organize events, and manage communication for a program serving preschoolers from low-income families

University Peer Mentor, BYU Office of First Year Experience June 2024 – February 2025

- Co-taught two sections of Introduction to University (UNIV 101) with faculty
- Mentored 45 first-year students through regular one-on-one meetings; completed QPR gatekeeper training

Full-Time Missionary, California Riverside Mission October 2021 – December 2023

The Church of Jesus Christ of Latter-day Saints

- Led and supported small groups of fellow volunteers
- Planned and gave 80+ trainings on teaching skills to groups of 8 to 100 volunteers; mentored four new volunteers one-on-one
- Worked in the mission office coordinating housing and travel logistics and managing technology

Squad Leader, Air Academy High School Marching Band, Colorado Springs, CO May 2020 – May 2021

- Led the trumpet section; the band won two state championships and marched in the 2018 Rose Parade

SERVICE

Volunteer, Global Early Detection Initiative, Utah Chapter (community outreach) — August 2026 – Present

Church Instructor, The Church of Jesus Christ of Latter-day Saints — August 2026 – Present

Congregation Activities Committee and Leadership Council — September 2025 – April 2026

Congregation Clerk (records) and Leadership Council — May 2024 – August 2024

Boy Scouts of America; Eagle Scout project for the National Mill Dog Rescue — 2014 – 2020

CERTIFICATIONS AND TRAINING

Thermo Fisher Scientific "Resolution Evolution" cryo-EM training (Vitrobot, Krios) — October 2026

Epic for Research Coordinators, UT Southwestern — May 2026

SCCR Research Ethics Seminar, BYU — April 2026

NHCO Certified Phlebotomy Technician (two-year certification, expired) — June 2024 – June 2026

QPR Suicide Prevention Gatekeeper Training, BYU — 2024

CONFERENCES ATTENDED

AI for Social Innovation Conference, BYU Ballard Center, Aspen Grove, UT — October 2026

Utah CryoEM Symposium, Brigham Young University — October 2026

American Chemical Society Spring Meeting, Atlanta, GA — March 2026

SKILLS

Laboratory: bacterial culture and plasmid work, SEM, ultra-high-vacuum systems, XPS, mass spectrometry, spin coating, flow reactor operation, Vitrobot, Krios cryo-TEM

Computational: Python, R, C#, Java, shell, Git and GitHub, ImageJ, SnapGene, Epic, cryo-ET processing software, DICOM and medical imaging pipelines, GPU workflows; AI coding agents since 2024, including agent skills and hooks

Communication: first author on three manuscripts and a co-author on several more; six research presentations and an invited guest talk; figure, poster, and interface design

ACTIVITIES

BYU Fencing Club — August 2026 – Present

Intramural Ultimate Frisbee, BYU — May 2025 – December 2025